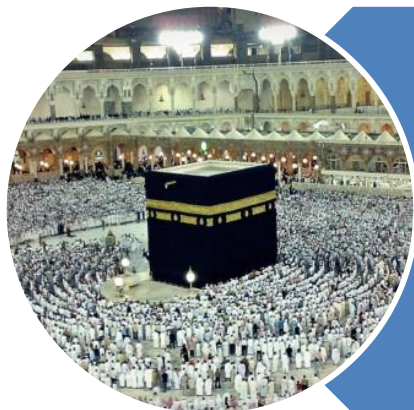


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Sorting Techniques

Introduction

C

P

C

S

2

0

4

3

Sorting - Introduction

- Fundamental problem in Computer Science
- Sorting is done to make searching easier
- Most programs do this:
 - Excel, Access, and others.



C

P

C

S

2

0

4

4

Introduction

- Techniques – Quadratic - $O(n^2)$
 - Selection Sort
 - Insertion Sort
 - Bubble Sort
- Techniques – Logarithmic - $O(n \log n)$
 - Merge Sort
 - Quick Sort

Quadratic Sorting

Selection Sort

C

P

C

S

2

0

4

6

Selection Sort - **Algorithm**

- 1) Find the **minimum/maximum** value in the list of n elements
 - Search from index 0 to index $n-1$
- 2) **Swap** that **minimum value** with the value in the **first position**
 - At index 0
- 3) **Repeat** steps 1 and 2 for the remainder of the list

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1								
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3	18	23	28	64	45	35	77	40
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3	18	23	28	64	45	35	77	40
Loop 4	18	23	28	35	45	64	77	40
Loop 5								
Loop 6								
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3	18	23	28	64	45	35	77	40
Loop 4	18	23	28	35	45	64	77	40
Loop 5	18	23	28	35	40	64	77	45
Loop 6								
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3	18	23	28	64	45	35	77	40
Loop 4	18	23	28	35	45	64	77	40
Loop 5	18	23	28	35	40	64	77	45
Loop 6	18	23	28	35	40	45	77	64
Loop 7								
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3	18	23	28	64	45	35	77	40
Loop 4	18	23	28	35	45	64	77	40
Loop 5	18	23	28	35	40	64	77	45
Loop 6	18	23	28	35	40	45	77	64
Loop 7	18	23	28	35	40	45	64	77
Final								

Selection Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	18	77	45	64	28	35	23	40
Loop 2	18	23	45	64	28	35	77	40
Loop 3	18	23	28	64	45	35	77	40
Loop 4	18	23	28	35	45	64	77	40
Loop 5	18	23	28	35	40	64	77	45
Loop 6	18	23	28	35	40	45	77	64
Loop 7	18	23	28	35	40	45	64	77
Final	18	23	28	35	40	45	64	77

C

P

C

S

2

0

4

Selection Sort - Analysis

- $T(n) = n + (n-1) + (n-2) + (n-3) + \dots + 3 + 2 + 1$
- $T(n) = n(n+1)/2$
- $T(n) = \frac{1}{2}n^2 + \frac{1}{2}n$
- $T(n) = O(n^2)$

Quadratic Sorting

Insertion Sort

Insertion Sort - **Algorithm**

- Playing Cards
 - Players usually keep cards in sorted order
 - When you pick up a new card
 - You make room for the new card and put into its proper place



C

P

C

S

2

0

4

Insertion Sort - **Algorithm**

- Descending Order
 - Starting with the 2nd element,
 - continually **SWAP** it with the previous element
 - Until
 - it found a bigger value **OR**
 - It reaches to the first index i.e. “0”

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1								
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2	77	45	23	64	28	35	18	40
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2	77	45	23	64	28	35	18	40
Loop 3	77	64	45	23	28	35	18	40
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2	77	45	23	64	28	35	18	40
Loop 3	77	64	45	23	28	35	18	40
Loop 4	77	64	45	28	23	35	18	40
Loop 5								
Loop 6								
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2	77	45	23	64	28	35	18	40
Loop 3	77	64	45	23	28	35	18	40
Loop 4	77	64	45	28	23	35	18	40
Loop 5	77	64	45	35	28	23	18	40
Loop 6								
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2	77	45	23	64	28	35	18	40
Loop 3	77	64	45	23	28	35	18	40
Loop 4	77	64	45	28	23	35	18	40
Loop 5	77	64	45	35	28	23	18	40
Loop 6	77	64	45	35	28	23	18	40
Loop 7								
Final	77	64	45	40	35	28	23	18

Insertion Sort

Index	0	1	2	3	4	5	6	7
Initial	23	77	45	64	28	35	18	40
Loop 1	77	23	45	64	28	35	18	40
Loop 2	77	45	23	64	28	35	18	40
Loop 3	77	64	45	23	28	35	18	40
Loop 4	77	64	45	28	23	35	18	40
Loop 5	77	64	45	35	28	23	18	40
Loop 6	77	64	45	35	28	23	18	40
Loop 7	77	64	45	40	35	28	23	18
Final	77	64	45	40	35	28	23	18

C

P

C

S

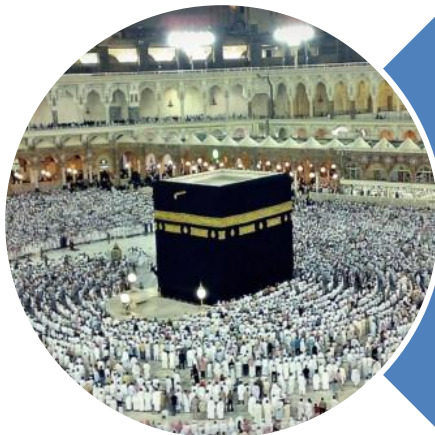
2

0

4

Insertion Sort - Analysis

- $T(n) = 1+2+3+ \dots + (n-3) + (n-2) + (n-1) + n$
- $T(n) = n(n+1)/2$
- $T(n) = \frac{1}{2}n^2 + \frac{1}{2}n$
- $T(n) = O(n^2)$



Allah (Alone) is Sufficient for us,
and He is the Best Disposer of
affairs (for us). (Quran 3 : 173)